EX NO: 4
DATE: 31/07/26.

SETTING UP A LOCAL AREA NETWORK (LAN)

AIM:

To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command and demonstrate resource / data sharing across the network.

PROCEDURE:

Step 1: Place the Devices.

* Open Cisco Packet Tracer
* From the device category panel, select End Devices and drag 4 PCs onto the workspace.
* Select Switches and drag 1 switch (e.g., 2960 series) onto the workspace.

Step 2: Cable the Devices.

* Click Connections → Select Copper Straight-Through Cable
* Click each PC, then click the switch, and select an available Fast Ethernet port to complete each connection.
* Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a successfull physical connection.

Step 3: Assign IP Addresses to PCs

* Click a PC → Desktop tab → IP Configuration.
* Enter the Static IP Address and Subnet Mask as per the IP table (section 5) for each PC.

Step 4: Configure the Switch Management IP

* Click the Switch → CLI tab.
* Enter the following Commands to assign a management IP:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5. 255.0.0.0
no shutdown
exit
```

Step 5: Verify Connectivity Using Ping.

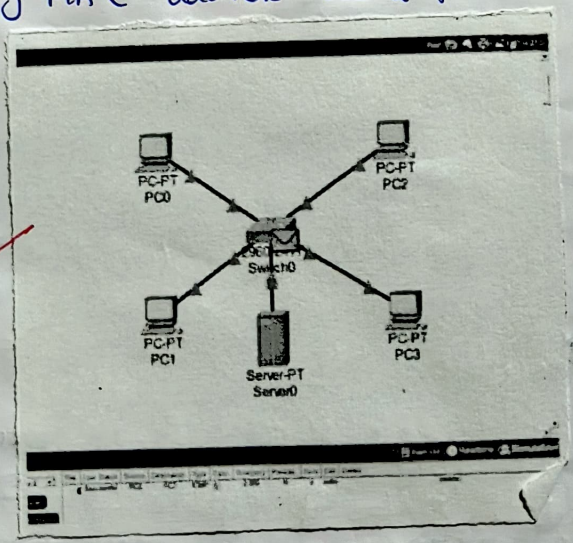
* Click a PC → Desktop tab → Command Prompt.
* Use the ping command to test connectivity to the other PCs, e.g. ping 10.1.1.2.
* Successful "Reply from..." message confirms connectivity.

Step 6: Demonstrate Data sharing

* Add a Server device (End devices → Server-PT) and connect it to the switch.
* Assign it an IP address via its Desktop → IP Configuration tab.
* Open the server's Services (Config) tab and enable the HTTP service.

Step 7: Observe Packet flow in Simulation Mode

* Switch Simulation mode (bottom-right of the workspace)
* Send the ping or web request between two devices and use the Play / Capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.



RESULT:

Thus the setup and Configuration of LAN using switch and Ethernet cables in your lab.